

RAJEE GANESAN

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PhD candidate in computational biology with 5+ years of experience designing data-driven research at scale. Track record of cross-institutional collaboration, quantitative analysis, and communicating complex findings to diverse stakeholders.

EDUCATION

Carnegie Mellon University

Ph.D. Candidate in Department of Biological Sciences

Eberly Center Future Faculty Program, Teaching Consultant Fellow

Pittsburgh, PA

April 2027

University of North Carolina, Chapel Hill

Bachelor of Science in Quantitative Biology, Minors in Data Science and Statistics; Cum Laude, Dean's List

Chapel Hill, NC

May 2022

EXPERIENCE

Carnegie Mellon University, Departments of Biological Science, Computational Biology

Doctoral Researcher in Andreas Pfenning Neurogenomics Lab

Pittsburgh, PA

July 2022 - Present

- Designed and executed large-scale computational genomics projects integrating machine learning, comparative genomics, and epigenomics across 280+ vertebrate genomes, delivering findings to interdisciplinary stakeholders from bench scientists to executive leadership
- Built and validated novel bioinformatics pipelines (SLURM HPC) processing terabyte-scale datasets; diagnosed and resolved critical production bugs, maintaining analytical integrity across a 577-genome project
- Collaborated with 10+ co-authors across 4 institutions; translated complex genomic analyses into clear narratives for diverse audiences including consortium updates, annual reviews, and peer-reviewed publications.
- Mentored 6 students (undergrad through PhD) on independent research projects; all placed into PhD or MD programs, including Best Honors Thesis, CMU School of Computer Science

Terrence Furey Lab, UNC Department of Medicine

Senior Research Intern

Chapel Hill, NC

Aug 2019 - Apr 2022

- Designed bioinformatics pipelines applying statistical modeling and allelic imbalance analysis to characterize genetic risk loci for Crohn's disease across large-scale cohort data
- Completed Honors Thesis resulting in peer-reviewed publication in Nature Communications; presented findings at state-level research symposium and developed expertise translating complex genomic results for clinical and non-specialist audiences

NIEHS/NIH, NIAMS/NIH Cold Spring Harbor Laboratory, La Jolla Institute for Allergy and Immunology

2019-2022

Research Intern & Fellow

- Completed four independent research internships across federal and academic institutions, contributing to projects in neutrophil biology, RNA structure, and immunology; co-authored publication in J Exp Med

SELECTED PUBLICATIONS

- Formenti G, [...] **Ganesan R**, [...], Jarvis ED, et al. (*in review*) The Vertebrate Genomes Project Phase I: A global reference genome resource.
- **Ganesan R**, Li X, Sestili H, Pfenning AR. (*in prep*) B cell enhancer evolution associated with mammalian social behavior.
- **Ganesan R**, [...] Jarvis ED, Pfenning AR, Vernes SC, VGP Vocal Learning Group. (*in prep*) VGP package: Shared genetic and neural features among humans and other vocal learning lineages.
- Marcus S, Davenport MH, Harter E, **Ganesan R**, Lee C, Phan BN, Terra N, Pfenning AR, Jarvis ED. (*in prep*) Transcriptomic signatures of long-range projection class neurons for vocal control.
- **Ganesan R**, Pfenning AR. (2025) Evolution of Mammalian Regulatory Networks in the Brain. *Annual Review of Animal Biosciences*. 14.
- Nishiyama NC, Silverstein S, Darlington K, Kennedy Ng MM, Clough KM, Bauer M, Beasley C, Bharadwaj A, **Ganesan R**, et al. (2026) eQTL in diseased colon tissue identifies novel target genes associated with IBD. *Nature Communications*.
- Marki A, Buscher K, Lorenzini C, Meyer M, Saigusa R, Fan Z, Yeh YT, Hartmann N, Dan JM, Kiosses WB, Golden GJ, **Ganesan R**, et al. (2021) Elongated neutrophil-derived structures are blood-borne microparticles formed by rolling neutrophils during sepsis. *J Exp Med*.

SELECTED ACHIEVEMENTS

Awards: Best Poster, CMU Department of Computational Biology Retreat (2026); Travel Award, CMU Mellon College of Sciences (2026); Margaret Carver Award (2025); Exemplary Teaching Assistant (2025), Carolina Research Scholar (2022)

Presentations: Selected talks at Evolution Conference, Genes Brain & Behavior Conference, and Vertebrate Genome Conference (2026); posters at Society for Neuroscience (2025), Vertebrate Genome Conference (2025)

Teaching: Instructor, NIH/FAES single-cell RNA-seq courses (BIOF555/556), 5 semesters; Teaching Assistant and Guest Instructor (CMU, Genetics, Neurobiology of Disease, Genomics and Epigenetics of the Brain)

Leadership: Data Analyst, UNC Division One Baseball; Science Columnist & Opinion Editor, UNC Daily Tar Heel

SKILLS

Computational: Python, R, Bash, Java, SAS; Seurat, Scanpy, GATK, Git/GitHub, SLURM, Jupyter, RMarkdown, Microsoft Office

Research Methods: Machine learning, large-scale data analysis, scientific writing, cross-functional collaboration, stakeholder presentation, research project design, team management and mentorship, AI literacy and tool integration